The Analytic Contraction Theorem

Kähler MMP

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Contents

1 Overview

The aim of this note is to introduce the contraction theorems in the Kähler minimal model program.

2 Fujiki's blowing-down theorem and the Grauert contraction theorem

Theorem 2.1. *Let X be a reduced complex space and A an effective* **Cartier** *divisor. Assume $f : A \to A'$ is a proper surjective morphism to another complex space A', such that*

1. *The restriction $\mathcal{O}_A(-A)$ is f-ample,*
2. *The higher direct image $R^1 f_* \mathcal{O}_X(-jA) = 0$ for all $j > 0$.*

Then there exists a contraction morphism $F : X \to X'$ such that $F|_A = f$. Assume $\mathcal{S}_{X,A,f}$ is defined by the exact sequence

$$0 \to \mathcal{S}_{X,A,f} \to \mathcal{O}_X \to \mathcal{O}_A/im(f^*\mathcal{O}_{A'} \to \mathcal{O}_A),$$

then $F_*(\mathcal{S}_{X,A,f}) = \mathcal{O}_X$.

3 Kollár-Mori's extension of contraction theorem

Theorem 3.1 ([KM92, Theorem 11.4]).

The proof of Kollár-Mori's Theorem 11.4 is very likely motivated by Horikawa's paper; the proof consists of two parts:

(1) First showing that the obstruction of deformation of the contraction morphism lies in $R^1 f_* \mathcal{O}_X$ (in particular the vanishing of the 1st direct image implies unobstructedness of the deformation),

(2) Second using the Douady space argument showing that extension from the formal neighborhood to an actual neighborhood is unconditional.

We will sketch the idea of the proof, and some issues for discussion are highlighted in red boxes.

3.1 Step 1. Infinitesimal level extension (Formal extension)

Kollár and Mori did not prove this part. Our proof is motivated by [DH24, Theorem 5.8], while Horikawa's approach uses some Čech process.

First, we have the following fundamental exact sequence

$$0 \to \mathcal{O}_X(-jX_0) \to \mathcal{O}_X \to \mathcal{O}_{jX_0} \to 0, \tag{1}$$

which will then induce

$$\boxed{0 \to \mathcal{O}_{X_0}(-jX_0) \to \mathcal{O}_{(j+1)X_0} \to \mathcal{O}_{jX_0} \to 0.} \tag{2}$$

Remark: For (2) to be exact, one requires X_0 to be Cartier (correct; the $\mathbb{Q}$-Cartier condition is insufficient)?

On the other hand, (1) holds as long as X_0 is a Weil divisor, and does not require any singularity assumption?

So that taking the direct image on (2) will induce a long exact sequence

$$0 \to f_*(\mathcal{O}_{X_0}(-jX_0)) \to f_*\mathcal{O}_{(j+1)X_0} \to f_*\mathcal{O}_{jX_0} \to R^1 f_*\mathcal{O}_{X_0}(-jX_0) \to R^1 f_*\mathcal{O}_{(j+1)X_0} \to \cdots, \quad (3)$$

Similarly to the proof of [DH24, Theorem 5.8], we need to show that certain 1st direct images vanish (for [DH24], the positivity and singularity conditions are needed in order to apply the Kawamata–Viehweg vanishing theorem). For our setting, we need to prove

$$R^1 f_*\mathcal{O}_{X_0}(-jX_0) = 0,$$

(which is expected to follow from the assumption $R^1 f_*\mathcal{O}_{X_0} = 0$; to do this, my idea is to apply the projection formula for higher direct images

$$R^1 f_*(\mathcal{O}_{X_0}(-jX_0)) = R^1 f_*(\mathcal{O}_{X_0} \otimes \mathcal{O}_X(-jX_0)) = R^1 f_*(\mathcal{O}_{X_0} \otimes f^*\mathcal{L}) = R^1 f_*\mathcal{O}_{X_0} \otimes \mathcal{L} = 0.$$

Here is the gap that I cannot fix.

QUESTION 1: How to show that

$$\mathcal{O}_X(-jX_0) = f^*\mathcal{L}?$$

After that we can inductively define the $(j+1)$-th order infinitesimal extension from the j-th order, by defining

$$Y_0^{(j+1)} = \mathrm{Specan}(f_*\mathcal{O}_{(j+1)X_0}),$$

which will induce the infinitesimal thickening of the morphism (since the construction here is functorial).

$$\begin{tikzcd} X_0^{(n+1)} \arrow[r, "f^{(n+1)}"] & Y_0^{(n+1)} \\ X_0^{(n)} \arrow[u, hook] \arrow[r, "f^{(n)}"] & Y_0^{(n)} \arrow[u, hook] \end{tikzcd}$$

3.2 Step 2. Douady space argument (Effectiveness)

From the formal neighborhood to an analytic neighborhood, one typically does not need to impose additional conditions. Our proof in this section is motivated by [**?**], rewritten in a clearer manner with the help of [BM19] (last chapter).

> **Remark 3.2.** I am trying to give a new proof that replaces the Douady space by the Chow–Barlet cycle space, as the cycle space is more powerful when dealing with non-reduced structure. The proof below is still in the Douady space setting.

The idea is to use the correspondence between a holomorphic morphism and its graph, so that it is possible to convert the problem of deforming a holomorphic map into a problem of deforming a complex subspace. Thus, the Douady space can be used to "parameterize" the family of holomorphic maps, and a holomorphic section of the relative Douady space **glues** fiberwise holomorphic maps together.

Let

$$\pi' : \mathcal{Y} \to \mathrm{Def}(Y_0)$$

be the Kuranishi family of deformations of Y_0 (here $\mathrm{Def}(Y_0)$ is some complex analytic space). It can be very singular, but there exists a holomorphic curve $\phi : U \to \mathrm{Def}(Y_0)$ passing through $0 \in \mathrm{Def}(Y_0)$, for some open $U \subset S$. Therefore, we have the following pullback diagram.

$$\begin{tikzcd} Y \arrow[r] \arrow[d, "h"] & \mathcal{Y} \arrow[d, "\pi'"] \\ U \arrow[r, "\phi"] & \mathrm{Def}(Y_0) \end{tikzcd}$$

Let $h \times g : Y \times X \to U \times U$ be the product family. We can define the relative Douady space $\mathcal{D} = \mathrm{Douady}(Y \times X / U \times U)$ that parametrizes the complex subspaces in the fibers of $h \times g$. Douady (and Pourcin for the relative version) proved that $\mathcal{D}$ admits a complex space structure, and there exists a holomorphic map $\Phi : \mathcal{D} \to U \times U$.

Furthermore, by the same proof that the Hom scheme is Zariski open in the Hilbert scheme, there exists an open subset $\mathcal{D}' \subset \mathcal{D}$ that parametrises the graphs of morphisms. More precisely, a point $[\Gamma_{s,t}] \in \mathcal{D}'$ represents a complex subspace $\Gamma_{s,t} \subset X_s \times Y_t$ such that the natural projection $q_1 : \Gamma_{s,t} \to X_s$ is an isomorphism.

$$\begin{tikzcd} & \Gamma_{s,t} \arrow[dl, "q_1"'] \arrow[d, hook] \arrow[dr, "q_2"] & \\ X_s & X_s \times Y_t \arrow[l] \arrow[r] & Y_t \end{tikzcd}$$

By definition, the representative $[\Gamma_f]$ of the graph of the morphism $f : X_0 \to Y_0$ lies in the fiber $\Phi^{-1}(0,0)$ and $[\Gamma_f] \in \mathcal{D}'$. The collection of infinitesimal thickenings of the morphism $\{f^{(n)}\}_{n\in\mathbb{N}}$ (constructed in Step 1) defines a formal section around $(0,0) \in U \times U$, as the picture below shows.

$$\begin{tikzcd}
\widetilde{y}_0 \in \widetilde{E}_0 \arrow[r, hook] \arrow[d, "p"] & \widetilde{S}_0 \arrow[r, hook] \arrow[d, "p"] & \widetilde{X} \arrow[d, "p"] \\
\widehat{y}_0 \in E_0 \arrow[r, hook] \arrow[d, "\pi"] & \widehat{S}_0 \arrow[r, hook] \arrow[d, "\pi"] & \widehat{X} \arrow[d, "\pi"] \\
y_0 \in C_0 \arrow[r, hook] & S_0 \arrow[r, hook] & X
\end{tikzcd}$$

Figure 1: Formal section of the relative Douady space.

Thus, by Artin [Art70], after further shrinking U, there exists a holomorphic section $\sigma : U \times U \to \mathcal{D}'$ on some analytic neighborhood of $0 \in U \times U$, and we restrict the section to the diagonal $\sigma : U \to \mathcal{D}'$.

Next, we claim that the holomorphic section $\sigma : U \to \mathcal{D}'$ defines a holomorphic map $X \to Y$ over U which is an extension of $f : X_0 \to Y_0$.

To see this, note that the section $\sigma : U \to \mathcal{D}'$ gives an analytic family of subspaces $(\Gamma_s)_{s\in U}$ in $X \times Y$. By [BM19, Theorem 4.3.3], the graph

$$G_U = \{(x, y, s) \in (X \times Y) \times U \mid (x, y) \in |\Gamma_s|\}$$

admits a complex space structure. (Barlet proves the result in the Barlet–Chow cycle space setting; it should be true for the Douady space as well.) This complex space structure is important, as it glues the fibers together.

Therefore, we have the natural projections (in the category of complex spaces) shown below.

$$\begin{tikzcd}
 & G_U = \bigcup_{s\in U} \Gamma_s \arrow[dl, "p_1"'] \arrow[dr, "p_2"] \arrow[dd, "p_3"] & \\
X \arrow[dr] & & Y \arrow[dl] \\
 & U &
\end{tikzcd}$$

Since, by the definition of $\mathcal{D}'$, the restriction $p_1|_{\Gamma_s}$ is an isomorphism for all $s \in U$, the map p_1 is an isomorphism and therefore there exists a holomorphic map

$$f = p_2 \circ p_1^{-1} : X \to Y$$

that makes the diagram commute, which completes the proof.

3.3 Step 3. Applications of Theorem 1 to MMP steps

We finally apply Theorem 1 to MMP steps; the proof is motivated by [**?**, Theorem 4.1].

We show that the following conditions hold (see Corollary 2.12 of the author's paper).

Theorem 3.3. *Assume that we have a contraction morphism $f : X_0 \to Y_0$ from a projective variety X_0 with canonical singularity.*
(1) If $R^1 f_*\mathcal{O}_X = 0$, then the extension $f : X \to Y$ is a contraction morphism (i.e., $f_*\mathcal{O}_X = \mathcal{O}_Y$).
(2) If $f : X_0 \to Y_0$ is a $(K_{X_0} + \Delta_0 + \beta_0)$-divisorial or flipping contraction (in an MMP step), then there exists an extension $f : X \to Y$ that is a fiberwise bimeromorphic contraction morphism.
(3) If on the central fiber $f : X_0 \to Y_0$ is a divisorial or fiber type contraction, the same holds for the nearby fibers; however, a flipping contraction on the central fiber may restrict to the identity on the nearby fibers (Totaro's example).

Proof of the contraction extending to a contraction morphism on some neighborhood of X_0. First, by the construction in Step 1, we know that on any n-th order infinitesimal thickening, we have that

$$\mathcal{O}_{Y,y} \to (f_*\mathcal{O}_X)_y$$

is an isomorphism. Then we can apply the formal function theorem so that

QUESTION 3: Why is the deformation of a contraction morphism still a contraction morphism?

On the infinitesimal level, we define the structure sheaf on the target to be a direct image, say

$$\mathcal{O}_{Y_0^{(j+1)}} := f_*\mathcal{O}_{(j+1)X_0};$$

however, it is not clear to me why the formal-to-actual extension is still a contraction morphism, thus I cannot actually prove Theorem 3(1).

QUESTION 4: (Relative MMP vs. Absolute MMP vs. fiberwise MMP)
Could I ask what the difference is between these three concepts?

QUESTION 5: (Analytic flip vs. Algebraic flip)
Could I ask about the new phenomena that appear in analytic flips? That is, an analytic flip may flip one component while an algebraic flip may flip several components.

Proof of fiberwise bimeromorphic. Taking (1) as given, we prove (2). By Garf-Kirschner's decomposition theorem, on a projective variety with rational singularity, we can represent $-(K_{X_0} + \Delta_0 + \beta_0)$ by some ample divisor $-(K_{X_0} + \Delta_0 + B_0)$. Therefore, by the relative Kawamata–Viehweg vanishing theorem, $R^1 f_* \mathcal{O}_{X_0} = 0$. Therefore, by Theorem 3.3, there exists some extension $f : X \to Y$. Since the deformation is a flat morphism, we have $\dim X = \dim Y$. Since a contraction in the MMP step preserves the normality condition, Y_0 is normal. Since being normal is an open condition, Y is still normal. Since we assume Theorem 3(1), we have that $f : X \to Y$ is a contraction morphism; therefore by Zariski's main theorem (or Nakayama II.2.12) we have that $f : X \to Y$ is actually a bimeromorphic morphism, and in particular it is fiberwise bimeromorphic.

Proof that the contraction type is preserved. The same as [**?**, Theorem 4.1], using semi-continuity. See also [KM92, Theorem 12.3.1]. We copy the statement of KM92 below.

Let $g : X \to S$ be a proper flat morphism of complex spaces. Assume that for some $0 \in S$ the fiber X_0 is a projective variety with only $\mathbb{Q}$-factorial rational singularities and $\dim X_0 \geq 3$. Let $f_0 : X_0 \to Y_0$ be the contraction of an extremal ray $C_0 \subset X_0$. By Theorem 3.3, there is a proper flat morphism $Y \to S$ and a factorisation

$$g : X \xrightarrow{f} Y \to S$$

Then there is an open neighborhood $0 \in U \subset S$ such that if f_0 contracts a subset of codimension at least two (resp. contracts a divisor; resp. is a fiber space of generic relative dimension k) then f_s contracts a subset of codimension at least two (which may be empty) (resp. contracts a divisor; resp. is a fiber space of generic relative dimension k) for all $s \in U$.

Proof. ∎

4 Das-Hacon's contraction theorem for generalized Kähler PLT pairs

In this section, we introduce the contraction theorem of Das and Hacon.

Theorem 4.1 ([DH24, Theorem 5.8]). *Let $(X, S + B + \boldsymbol{\beta})$ be a generalized PLT pair with $\lfloor S + B \rfloor = S$ irreducible, such that the following conditions hold:*

1. *S is a $\mathbb{Q}$-Cartier divisor,*
2. *There exists a contraction morphism $f : S \to T$ such that $-S|_S$ is f-ample,*
3. *The restriction $-(K_X + S + B + \boldsymbol{\beta})|_S$ is Kähler over T.*

Then we can find a (bimeromorphic) contraction morphism $F : X \to Y$ with $F|_S = f$.

Remark 4.2. We briefly sketch the idea of the proof. Compared with the Fujiki blowing-down theorem, we do not have the Cartier condition on S and we do not have the vanishing of the higher direct image (of the conormal sheaf) in the statement. Since S is $\mathbb{Q}$-Cartier, there exists $r \in \mathbb{Z}$ such that rS is Cartier. We first show that there exists an extension on the infinitesimal thickening rS, with positivities preserved under thickening. Second, we apply Serre vanishing and a change-of-index trick to show that the higher direct images of the conormal sheaf vanish. Then applying the Fujiki blowing-down theorem yields the result. The main difficulty of the proof lies in showing that the obstruction to the infinitesimal extension vanishes. To do this, we need adjunction for the generalized PLT pairs and the Kawamata–Viehweg vanishing theorem for complex analytic spaces.

Proof. Assume $r \in \mathbb{Z}_+$ to be an integer such that rS is a Cartier divisor. We have the following short exact sequence of sheaves of $\mathcal{O}_X$-modules for the thickening of S:

$$0 \to \mathcal{O}_S(-jS) \to \mathcal{O}_{(j+1)S} \to \mathcal{O}_{jS} \to 0.$$

On the other hand, we have the quotient exact sequence

$$0 \to \mathcal{O}_X(-(j+1)S) \to \mathcal{O}_X(-jS) \to \mathcal{E}_j \to 0,$$

where $\mathcal{E}_j$ is a rank 1 reflexive sheaf supported on S.
Combining these two exact sequences, we get ■

5 Applications of Analytic Contraction Theorems

5.1 Applications in Kähler minimal model program

Das, Hacon, and Păun use Theorem 3.3 to prove the semi-stable MMP for Kähler 4-folds over a curve.

Theorem 5.1 ([DHP24, Theorem 8.12])**.** *Let $f : (X, B) \to T$ be a semi-stable klt pair of dimension 4 and $W \subset T$ a compact subset. If $(X/T; W)$ is $\mathbb{Q}$-factorial and $K_X + B$ is effective over W, then we can run the $(K_X + B)$-MMP over a neighborhood of W in T which ends with a minimal model over W.*

5.2 Applications in algebraic approximation problems

Lin uses Theorem 3.3 in the proof of the following result.

Theorem 5.2 ([?, Proposition 1.7])**.** *Let X' be a normal compact complex variety and let X be a compact complex variety with at worst rational singularities that is bimeromorphic to X'. Assume that X' has a Y-locally trivial algebraic approximation for every subvariety $Y \subset X'$ satisfying* $\dim Y \leq \dim X' - 2$*. Then X has an algebraic approximation.*

Remark 5.3. Without going into too much detail, the Y-locally trivial condition (for $\dim Y \leq \dim X' - 2$) in the theorem implies that an algebraic approximation of X' can induce an algebraic approximation of the higher model X'', provided that the bimeromorphic map

$$X'' \to X'$$

is an isomorphism in codimension 1 on the target.

Proof. Let

$$\tau : X' \dashrightarrow X$$

be the bimeromorphic map in the assumption. The idea is to take the common resolution

$$\begin{tikzcd} & \tilde{X} \arrow[dl, "\nu"'] \arrow[dr, "\eta"] & \\ X' \arrow[rr, dashed, "\tau"] & & X \end{tikzcd}$$

We then try to apply the target-stable result on ν and the source-stable result on η (that is, a deformation of X induces a deformation of ν, and a deformation of $\tilde{X}$ induces a deformation of η).
By [?, Lemma 2.7], an algebraic approximation on X' will induce an algebraic approximation on $\tilde{X}$.
Since the target X has rational singularities, the direct image $R^1\eta_*\mathcal{O}_{\tilde{X}} = 0$. Therefore, by Theorem 3.3, the algebraic approximation on $\tilde{X}$ descends to an algebraic approximation on X. ∎

5.3 Applications in deformation of Calabi-Yau problem

Kollár uses Theorem 3.3 to prove the following result (which says that a contraction always extends, except when the target is uniruled or an abelian variety).

Theorem 5.4 ([?, Theorem 33])**.** *Let X be a projective variety with rational singularities, Y a normal variety, and $g : X \to Y$ a surjective morphism with connected fibers. Assume that Y is not uniruled. Then at least one of the following holds:*

(1) Every small deformation of X gives a deformation of $(g : X \to Y)$.
(2) There is a quasi-étale cover $\tilde{Y} \to Y$, a normal variety Z, and positive dimensional abelian varieties A_1, A_2 such that the lifted morphism $\tilde{g} : \tilde{X} := X \times_Y \tilde{Y} \to \tilde{Y}$ factors as

$$\begin{tikzcd} \tilde{X} \arrow[r] \arrow[d, "\tilde{g}"'] & Z \times A_1 \times A_2 \arrow[d] \\ \tilde{Y} \arrow[r, "\cong"'] & Z \times A_1 \end{tikzcd}$$

Kollár expects that Theorem 3.3 is useful in proving the following conjecture about deformation of Calabi–Yau fibrations.

Conjecture 5.5 ([?, Theorem 52]). *Let $g_0 : (X_0, \Delta_0) \to B_0$ be a relatively minimal log Calabi–Yau fiber space where (X_0, Δ_0) is a proper klt pair and $H^2\left(X_0, \mathcal{O}_{X_0}\right) = 0$.*
Let (X, Δ) be a klt pair and $h : (X, \Delta) \to (0 \in S)$ a flat proper morphism whose central fiber is (X_0, Δ_0).
Then, after passing to an analytic or étale neighborhood of $0 \in S$, there is a proper, flat morphism $B \to (0 \in S)$ whose central fiber is B_0 such that g_0 extends to a log Calabi–Yau fiber space $g : (X, \Delta) \to B$.

5.4 Applications in the invariance of plurigenera problems

The work of Levine, Cao, and Păun on invariance of plurigenera adopts a very similar strategy to Theorem 3.3. They first prove that infinitesimal extensions of the pluricanonical section exist when a certain $\bar{\partial}$-equation holds and satisfies some L^2-estimate.

Theorem 5.6. *Let $f : X \to \Delta$ be a smooth family of Kähler manifolds,*

$$s \in H^0(X_0^{(k)}, \mathcal{L}|_{X_0^{(k)}})$$

with $\mathcal{L} = (m-1)K_X$, which admits a C^∞ extension s_k so that if we write $\bar{\partial} s_k = t^{k+1}\Lambda_k$, the integral

$$\int_X \left|\frac{\Lambda_k}{dt}\right|^2 e^{-(1-\varepsilon)\varphi_L} dV < \infty,$$

converges for any $\varepsilon > 0$. Then there exists a section $\widehat{s}$ of $\mathcal{L}|_{X_0^{(k+1)}}$ such that $s = \pi_k(\widehat{s})$.

Remark 5.7. Compared with the 1st direct image vanishing condition in Theorem 3.3, they impose an L^2 condition. (I am not entirely sure whether this is a parallel condition.)

Remark 5.8. Levine proved a similar result: a smooth pluricanonical section on the central fiber admits an infinitesimal thickening. Cao and Păun gave an alternative proof of Levine's statement.

The extension of the pluricanonical section from the formal neighborhood of the central fiber to an actual analytic neighborhood is unconditional (which is expected to be true, compared with our proof of Theorem 3.3 in Step 2).

Theorem 5.9. *Let $f : X \to \Delta$ be a smooth family of compact complex manifolds. Assume that the pluricanonical section on X_0 admits an infinitesimal extension; then it admits an extension on some analytic neighborhood of* 0*.*

The next topic we will discuss is positivities in Kähler families.

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